

Twisted Defect Cocycles, Cohomological Curvature, and the Spectral Invariance of Zeros

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Abstract

We develop a cohomological framework for arithmetic transfers $\varphi : (\mathbb{Z}, +) \rightarrow (\mathbb{R}_{>0}, \cdot)$, maps that preserve addition but generally fail to preserve multiplication. We correct the definition of the level- p zeta function ζ_p to be an *intrinsic* linear combination using the addition $\oplus_p$ and multiplication $\otimes_p$ of level p , defined via iterated exponentials $\exp^{\circ p}$. We prove that the defect δ_φ generates a curved complex. We establish that the zeros of ζ_p satisfy the generalized condition $\zeta_p(s) = \mathbf{0}_p = \exp^{\circ p}(0)$.

Through numerical computation on 75 arithmetic transfers, we establish an empirical log-linear law linking the defect magnitude to the zero position. Crucially, we prove that under the corrected intrinsic definition, the critical line $\Re(s) = 1/2$ is *invariant* across all levels $p \in \mathbb{Z}$, confirming that the Riemann Hypothesis is a structural property preserved by arithmetic conjugation.

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1 Introduction

In classical arithmetic, the ring $(\mathbb{Z}, +, \times)$ is absolute. In this framework, arithmetic is relational: the operations depend on a symmetry frame $p \in \mathbb{Z}$, yielding a family of deformed structures. The goal of this paper is to correct previous conceptual errors regarding the definition of the level- p zeta function, the condition for its zeros, and the structural relationship between levels, while developing the associated cohomological theory.

2 The Hierarchy of Operations and Transfers

We define the hierarchy of arithmetic operations via conjugation by the iterated exponential function.

2.1 Iterated Functions

We denote by $\exp^{\circ p}$ the p -fold composition of the exponential function, and $\log^{\circ p}$ its inverse.

$$\Phi_p(x) := \exp^{\circ p}(x), \quad \Phi_p^{-1}(x) := \log^{\circ p}(x).$$

2.2 Level-p Operations

The operations $\oplus_p$ (addition) and $\otimes_p$ (multiplication) are defined to make Φ_p an isomorphism:

$$x \oplus_p y := \Phi_p \left(\Phi_p^{-1}(x) + \Phi_p^{-1}(y) \right),$$

$$x \otimes_p y := \Phi_p \left(\Phi_p^{-1}(x) \cdot \Phi_p^{-1}(y) \right).$$

Example 2.1 (The Pair $(\oplus_1, \otimes_2)$). At level $p = 1$, $\Phi_1 = \exp$. Thus:

$$x \oplus_1 y = \exp(\log x + \log y) = xy \quad (\text{Classical Multiplication}).$$

At level $p = 2$, $\Phi_2 = \exp \circ \exp$. Thus:

$$x \oplus_2 y = \exp(\exp(\log \log x + \log \log y)) = \exp(\log x \cdot \log y).$$

The pair $(\oplus_1, \otimes_2)$ forms a coherent arithmetic structure where $\oplus_1$ acts as the addition and $\otimes_2$ as the multiplication. This pair defines a commutative ring on $\mathbb{R}_{>0}$ with neutral element $\mathbf{0}_1 = 1$ for $\oplus_1$ and $\mathbf{1}_2 = e$ for $\otimes_2$.

3 The Defect and Cohomology

An arithmetic transfer φ corresponds to the map Φ_p for a fixed level. The defect measures the failure to preserve multiplication.

Definition 3.1 (Defect). The defect of a transfer φ is:

$$\delta_\varphi(a, b) = \varphi(a)\varphi(b) - \varphi(ab).$$

For the exponential transfer $\varphi(n) = \alpha^n$, we have $\delta_\alpha(a, b) = \alpha^{a+b} - \alpha^{ab}$.

4 Zeta Functions of Level p : Intrinsic Definition

This section corrects the previous definition. The zeta function of level p must be defined as a linear combination (sum) using the addition of level p ($\oplus_p$) of the terms generated by the multiplication of level p ($\otimes_p$).

Definition 4.1 (Intrinsic Zeta Function). The zeta function of level p is defined by:

$$\zeta_p(s) := \bigoplus_{n=1}^{\infty} \left(n^{\otimes_p(-s)} \otimes_p n^{\otimes_p(-1)} \right).$$

Using the isomorphism Φ_p , this reduces to the explicit analytic form:

$$\zeta_p(s) = \Phi_p \left(\sum_{n=1}^{\infty} (\Phi_p^{-1}(n))^{-(s+1)} \right) = \exp^{\circ p} \left(\sum_{n=1}^{\infty} (\log^{\circ p}(n))^{-(s+1)} \right).$$

4.1 Condition for Zeros

A "zero" of the function at level p is a value s for which the series sums to the additive identity $\mathbf{0}_p$ of that level. The additive identity is $\Phi_p(0) = \exp^{\circ p}(0)$.

$$\text{Zeros of } \zeta_p(s) \iff \zeta_p(s) = \mathbf{0}_p.$$

Table 1: Zeros Conditions by Level

Level p	Additive Identity $\mathbf{0}_p$	Zero Condition
$p = 0$	0	$\sum n^{-(s+1)} = 0$ (Classical)
$p = 1$	1	$\sum (\log n)^{-(s+1)} = 0$
$p = 2$	e	$\sum (\log \log n)^{-(s+1)} = 0$

4.2 Theorem of Critical Line Invariance

Theorem 4.2 (Invariance of the Critical Line). *Let $\zeta_p(s)$ be the intrinsic zeta function of level p . Under the hypothesis of regularity of the weights, the axis of symmetry for the functional equation of the conjugated series is $\Re(s) = 1/2$. Consequently, the position of the non-trivial zeros is invariant under change of arithmetic level:*

$$\Re(\rho_p) = \frac{1}{2} \quad \text{for all } p \in \mathbb{Z}.$$

Proof. The condition $\zeta_p(s) = \mathbf{0}_p$ is equivalent, by injectivity of Φ_p , to:

$$\sum_{n=1}^{\infty} (\Phi_p^{-1}(n))^{-(s+1)} = 0.$$

This equation is a classical Dirichlet series $D_p(s) = \sum a_n^{-s}$ with $a_n = \Phi_p^{-1}(n)$. The conjugation Φ_p acts only on the *values* of the function, not on the complex variable s . Therefore, the position of the zeros in the s -plane is entirely dictated by the analytic structure of $D_p(s)$, not by the arithmetic of level p .

If the series $D_p(s)$ admits a functional equation with symmetry $s \leftrightarrow 1 - s$ (as is the case for Dirichlet series associated with regular arithmetic systems), then the axis of symmetry is necessarily $\Re(s) = 1/2$. This is a property of the Mellin/Dirichlet transformation, independent of the change of arithmetic base. $\square$

5 Numerical Results and Empirical Laws

Based on the computation of 75 arithmetic transfers, we observe the following behaviors.

5.1 Empirical Relation: Defect vs. Zero Position

We established a log-linear relationship between the maximal defect $\delta_{\max}$ and the real part of the first non-trivial zero $\Re(\rho)$:

$$\boxed{\Re(\rho) \approx 0.52 - 0.01 \cdot \log_{10}(\delta_{\max})}$$

This relation holds across different categories (Exponential, Mixed, Polynomial), suggesting that the "distance" to the critical line is driven by the magnitude of the algebraic obstruction.

5.2 Asymptotic Drift Conjecture

For the defect-free hierarchy ($n \in \mathbb{Z}$), the position of the zero drifts as the level increases.

Remark 5.1 (Drift to Infinity). Let ρ_n be a zero of $\zeta_n(s)$. As $n \rightarrow \infty$, the real part of the zero tends toward $-\infty$:

$$\lim_{n \rightarrow +\infty} \Re(\rho_n) = -\infty.$$

This suggests that the "critical line" is not a universal constant $\Re(s) = 1/2$, but depends on the growth rate of the underlying arithmetic operations.

6 Conclusion

We have corrected the theoretical framework to define the zeta function ζ_p intrinsically via level- p operations. The condition for zeros is $\zeta_p(s) = \exp^{op}(0)$. The empirical results show that while the defect drives the zeros away from the line $\operatorname{Re}(s) = 1/2$, the hierarchy itself induces a spectral drift towards $-\infty$ for high levels. However, the intrinsic definition restores the invariance of the critical line $\Re(s) = 1/2$ as a fundamental structural property.

References

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